Science Lead guide

Our ecosystem runs on hydrogen, with no oxygen and no sunlight anywhere. The science lead turns the sections below into report sections 4a, 4b and 4c plus the rejected-alternative paragraph. The text in italics is ready to paste.

Locked decisions

These are settled, so the report, the site and the video must all say the same thing.

Decision	What we chose	Why	Source
Energy source	Hydrogen from seawater reacting with rock (serpentinisation)	The fuel Cassini actually measured at Enceladus; it needs no sunlight and no oxygen	Waite 2017; Kelley 2005
Base of the food web	Methane-making microbes in the warm pores of the chimneys, where vent hydrogen meets ocean CO_2	Lost City's methane microbes live in warm chimney interiors	Kelley 2005; Taubner 2018
Oxygen	**None anywhere**	No sunlight means no photosynthesis, so no oxygen. This keeps our argument against sulfide vents intact	Reasoning from the NOAA fact sheet
Food web	3 levels: methane-makers → single-celled grazers → Shadowfin (under 1 mm)	Only tiny organisms are known to live their whole life without oxygen	Danovaro 2010; Pauly & Christensen 1995
Habitat zone	One vent field of white carbonate chimneys, 50 km down (20 km of ice + 30 km of ocean)	Modelled on Lost City and Enceladus	Kelley 2005; NASA; Thomas 2016
Numbers on the gauges	Exact only where we calculate them (depth, pressure); ranges or plain words elsewhere	A precise number with no source invites "where is this from?"	—
Moon's name	Pyrothalasa (spelling to confirm)	Already on the live site	—

4a · Habitat and conditions

The brief asks for at least 4 conditions; we give 9, and each is tagged as evidence, calculation or assumption.

Condition	Our value	Evidence or assumption
Depth	50 km below the surface: 20 km of ice + a 30 km ocean	Based on Enceladus (NASA; Thomas 2016). The exact numbers are our choice
Pressure	about 56 bar at the seafloor, the same as ~550 m deep on Earth	Calculated: density × gravity × depth, with gravity from JPL (0.113 m/s^2)
Temperature	open ocean near freezing (~0 °C); vent fluid <40 to 90 °C; garden water ~8 °C	Vent fluid: Kelley 2005. Ocean near freezing: it sits under ice. **8 °C is an assumption**
Chemicals	hydrogen (H_2), carbon dioxide (CO_2), methane (CH_4)	Waite 2017 (plume); Kelley 2005 (Lost City)
Acidity	very alkaline, pH 9–11	Lost City's vent fluids (Kelley 2005)
Salinity	salty	Sodium salts in the plume's ice grains (Postberg 2009)
Oxygen	**none**	**Assumption:** no sunlight means no photosynthesis, the source of Earth's ocean oxygen
Light	none. Sunlight at Saturn is ~1% of Earth's, and it can't get through 20 km of ice	NASA Saturn facts + the inverse-square law
Food from above	none	On Earth, deep-sea life relies on food sinking from the sunlit surface (NOAA). Our surface has no life, so nothing sinks

Ready-to-use paragraph (about 90 words):

> *The Hydrogen Garden is a single field of white chimneys on the seafloor of Pyrothalasa, our fictional icy moon. It lies 50 km down, under 20 km of ice and 30 km of salty ocean, where the pressure is about 56 bar. No sunlight reaches it, so there is no photosynthesis and no oxygen. Warm, very alkaline fluid rich in hydrogen seeps from the chimneys into water that is otherwise near freezing. We modelled the garden on Lost City, a real vent field in the Atlantic, and the ocean on Saturn's moon Enceladus.*

4b · The energy source

As seawater soaks through rock rich in olivine, the two react (serpentinisation) and release hydrogen gas. Methane-making microbes then combine that hydrogen with carbon dioxide and use the energy released to grow:

$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$

Write the equation as plain text like this. A formula block shows up as raw code in the PDF export.

- **The strongest evidence, use it:** Cassini's hydrogen measurement shows that Enceladus' ocean chemistry favours making methane from CO_2 (Waite 2017). That links our microbes to Enceladus itself, not only to Lost City. It answers guiding question 2.
- **Where else we've seen it:** at Lost City, seawater reacting with mantle rock makes hydrogen- and methane-rich fluid, and methane-cycling microbes live inside the warm chimneys (Kelley 2005). An Earth methane-maker also grew in the lab under Enceladus-like conditions (Taubner 2018).
- **Why it lasts:** tidal heating from Saturn keeps the rocky core warm, and models suggest hydrothermal activity could continue for tens of millions to billions of years (Choblet 2017, still to check). It answers guiding question 1.
- **Uncertainty to name:** at Lost City the carbon comes from the rock, not from seawater CO_2 (Proskurowski 2008), so we assume enough CO_2 reaches the chimney pores.

Ready-to-use paragraph (about 110 words):

> *Our garden runs on hydrogen. When seawater soaks into rock rich in the mineral olivine, the two react and release hydrogen gas, a process called serpentinisation. Microbes called methanogens combine that hydrogen with carbon dioxide to make methane, and they use the energy released to grow ($CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$). This needs no sunlight and no oxygen. It happens today at Lost City on Earth, and Cassini found hydrogen in the plume of Saturn's moon Enceladus, at levels that favour exactly this reaction. Heat from Saturn's tides could keep the rock reacting for millions of years, long enough for an ecosystem to depend on it.*

4c · Design choices linked to evidence

This is the table the brief asks for. The "What we assume" column shows judges we know where the evidence stops. Use at least the first 3 rows; add more if page 2 has room.

#	Design choice	Real-world evidence	What we assume	Source
1	Energy comes from hydrogen made when water reacts with rock	Lost City's vents are driven by this reaction; Cassini found hydrogen in Enceladus' plume	Water keeps circulating through reactive rock on our moon	Kelley 2005; Waite 2017
2	Methane-making microbes form the base of the food web	Enceladus' ocean chemistry favours making methane from CO_2; methane microbes live in Lost City's chimneys; an Earth methane-maker grew under Enceladus-like conditions	Such microbes exist on our moon (habitable does not mean inhabited), and enough CO_2 reaches them	Waite 2017; Kelley 2005; Taubner 2018
3	The habitat is a vent field in a dark, salty, global ocean under ice	Enceladus has a global ocean, salts in its ice grains, and silica grains that point to hot water reacting with rock	Our moon is built like Enceladus	Thomas 2016; Postberg 2009; Hsu 2015
4	The whole food web lives without oxygen	Single-celled organisms and tiny animals (Loricifera) live their whole life without oxygen in a Mediterranean basin	Life like this could also evolve elsewhere. It is one study from one place	Danovaro 2010
5	The food web is short and its top predator is microscopic	Only ~10% of energy passes from one level to the next	Life without oxygen passes on even less (still needs a source)	Pauly & Christensen 1995
6	The energy supply lasts long enough for an ecosystem	Tidal heating can power long-lived hydrothermal activity at Enceladus	Our moon's core has been active just as long	Choblet 2017

Rejected alternatives (creativity, worth up to 20 points)

The rubric's top band needs at least one alternative compared. We have two, and the second shows we tested our own design.

Alternative	What it is	Why we rejected it	Source
Copy Earth's black-smoker vents	Tubeworms living on bacteria that use hydrogen sulfide	That reaction needs oxygen ($CO_2 + 4H_2S + O_2 \rightarrow CH_2O + 4S + 3H_2O$), and Earth's deep-sea oxygen is made by sunlight. These "sunless" vents still depend on the Sun	NOAA fact sheet; NOAA poster
Let a little oxygen seep down from the surface, so bigger animals can breathe	At Europa, Juno saw the surface ice being split into oxygen, a possible supply for its ocean	It borrows from a different moon and needs oxygen to cross 20 km of ice. If oxygen were there, the sulfide pathway would work too, which undercuts our first argument	Szalay 2024

We also rejected keeping hand-sized animals that live without oxygen: every animal known to live without oxygen is tiny, so it would be an unsupported claim.

Ready-to-use paragraph (about 100 words):

> *We first considered copying Earth's famous black-smoker vents, where giant tubeworms live on bacteria that use hydrogen sulfide. But that reaction needs oxygen, and the oxygen in Earth's deep sea is made by photosynthesis at the sunlit surface, so even these "sunless" vents depend on the Sun. We then considered letting a little oxygen seep down from the surface, as may happen on Europa, so that larger animals could breathe. We rejected that too: it is uncertain, and it would weaken our first argument. Instead our whole food web runs without oxygen, so its life stays microscopic.*

Hand-offs from the science lead

Send each teammate exactly what they need, so every part of the submission says the same thing.

To	Send	So they can
P2 · Ecology & Story	The 3 levels and who eats whom; the "no oxygen, so everything is microscopic" reason; what happens when the hydrogen stops	Write section 3, 4d and the video script
P3 · Evidence & Critic	The 4c table, the open checks below, and two limits: whether enough CO_2 reaches the pores, and that oxygen-free animals come from one study	Check the sources and write section 6
P3 (validation)	The lab test: a lab grows the Taubner 2018 methane-maker at high, medium and low hydrogen, keeping pressure, temperature and pH the same, and measures methane and cell numbers. Success = both fall as hydrogen falls	Fill in the what / who / how / success table
P5 · Artist	Single-celled grazers with cilia and a Shadowfin under 1 mm (spiny head, vase-shaped shell), drawn inside a "microscope lens"	Redraw the creatures for the report and the site
P6 · Web and P4 · Video	The final condition labels from 4a and the ready-to-use paragraphs	Keep the site's cards and the narration word-for-word consistent with the report

Still to check before it goes in the report

- [ ] Choblet 2017: confirm "tens of millions to billions of years" in the paper (its abstract wasn't available to check)
- [ ] Find a source for "life without oxygen gets less energy from its food" (try Fenchel & Finlay 1990, or a biology textbook)
- [ ] Confirm Loricifera are under 1 mm (Danovaro 2010 full text, or a zoology source)

- [ ] Confirm Enceladus' ~252 km radius on the JPL page, so the gravity number is fully sourced
- [ ] Re-export the PDF after these changes; the last export was missing "no oxygen"

Sources for the science sections

Every title and DOI has been checked. Abstracts were read for Waite, Kelley, Danovaro, Proskurowski and Szalay; P3 should read the rest before the report quotes them.

Source	Supports	Type
Waite et al. (2017), *Science* 356, 155–159	Hydrogen in the plume; the ocean's chemistry favours making methane from CO_2	Peer-reviewed
Kelley et al. (2005), *Science* 307, 1428–1434	Lost City: fluids <40 to 90 °C, pH 9–11, chimneys 30–60 m, methane microbes in warm interiors	Peer-reviewed
Taubner et al. (2018), *Nature Communications* 9, 748	An Earth methane-maker grew under Enceladus-like conditions	Peer-reviewed
Danovaro et al. (2010), *BMC Biology* 8, 30	Single-celled organisms and Loricifera living without oxygen	Peer-reviewed
Proskurowski et al. (2008), *Science* 319, 604–607	Lost City's carbon comes from rock, not seawater CO_2	Peer-reviewed
Thomas et al. (2016), *Icarus* 264, 37–47	Enceladus has a global ocean	Peer-reviewed
Postberg et al. (2009), *Nature* 459, 1098–1101	The ocean is salty	Peer-reviewed
Hsu et al. (2015), *Nature* 519, 207–210	Hot water reacting with rock	Peer-reviewed
Choblet et al. (2017), *Nature Astronomy* 1, 841–847	Long-lived heating (to check)	Peer-reviewed
Pauly & Christensen (1995), *Nature* 374, 255–257	~10% energy passed between levels	Peer-reviewed
Szalay et al. (2024), *Nature Astronomy*	Oxygen made at Europa's surface (only for the rejected alternative)	Peer-reviewed
NOAA chemosynthesis fact sheet	The sulfide reaction needs oxygen	Agency

Source	Supports	Type
NASA/JPL satellite parameters	Enceladus gravity	Agency